QUICK REFERENCE

ECG Cheat Sheet

Read any 12-lead ECG in 6 steps, the same way every time.

Dr. RRP CARDIO

Learn. Understand. Apply.

1 Set up and check

- Standard: **25 mm/s** and **10 mm/mV**.
- 1 small box = **0.04 s**. 1 large box = **0.2 s**.
- Right patient, right date, clean tracing, no tremor.
- Limb-lead reversal? Inverted P, QRS and T in lead I with an upright aVR.

2 Rate

- Regular: **300 ÷ large boxes** between two R waves.
- Irregular: QRS count in a **6-second strip × 10**.
- Normal 60–100 bpm. Below 60 brady, above 100 tachy.

Boxes (R–R)	1	2	3	4	5	6
bpm	**300**	**150**	**100**	**75**	**60**	**50**

3 Rhythm

- A P wave before **every** QRS, and a QRS after every P?
- Regular or irregular? Check R–R and P–P.
- **Sinus:** P upright in I, II, aVF and inverted in aVR, with a constant PR.
- QRS **narrow** (under 120 ms) or **wide**?
- No P waves: think AF, junctional or paced rhythm.

4 Axis normal −30° to +90°

Lead I	aVF	Axis
Positive	Positive	**Normal**
Positive	Negative	Left (LAD). Check lead II.
Negative	Positive	Right (RAD)
Negative	Negative	Extreme (northwest)

LAD: LAFB, inferior MI. RAD: RVH, PE, lateral MI.

5 Intervals

Interval	Normal
PR	120–200 ms (3–5 small boxes)
QRS	Under 120 ms (under 3 small boxes)
QTc	Under 450 ms men, under 460 ms women. Over 500 ms is high risk.

Bazett: QTc = QT ÷ √(R–R in seconds).

6 P, QRS, ST, T

- **P:** tall (over 2.5 mm) suggests right atrial enlargement. Wide or notched (120 ms or more) suggests left atrial.
- **Q:** pathological if 40 ms or wider, or over 25% of the R wave.
- **R progression:** R grows from V1 to V5. Transition at V3–V4.
- **LVH:** S in V1 + R in V5 or V6 over 35 mm.
- **ST:** measure at the J point against the TP or PR baseline.
- **T:** normally inverted in aVR and V1. Tall and peaked: hyperK or hyperacute MI.

Then ask — Is it normal? | Is it dangerous? | Compare with the old ECG

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Patterns to recognise

STEMI: which territory?

Territory	Leads	Artery
Septal	V1–V2	LAD
Anterior	V3–V4	LAD
Lateral	I, aVL, V5–V6	LCx, diagonal
Inferior	II, III, aVF	RCA, LCx
Posterior	ST↓ V1–V3, tall R. Add V7–V9.	RCA, LCx
Right ventricle	ST↑ in V4R	Proximal RCA

Inferior: ST↑ III over II points to RCA. II equal to or above III points to LCx. In RV infarct, avoid preload-lowering drugs such as nitrates.

STEMI criteria: new J-point ST↑ in 2 contiguous leads. 1 mm or more in all leads except V2–V3. In V2–V3: 2 mm men aged 40 and over, 2.5 mm men under 40, 1.5 mm women.

AV blocks

Type	What you see
1st degree	PR over 200 ms. Every P conducts.
Mobitz I (Wenckebach)	PR lengthens, then a beat drops. Grouped beating.
Mobitz II	Constant PR, sudden dropped QRS. Can progress: high risk.
3rd degree	P and QRS independent (AV dissociation). Slow, regular escape rhythm.

Bundle branch blocks QRS 120 ms or more

Block	V1	V6 (and I)
LBBB	Deep S or QS (the "W")	Broad, notched R (the "M"), no septal q
RBBB	rSR′ (the "M")	Wide S (the "W")

Chest pain with LBBB: compare with an old ECG and apply the Sgarbossa criteria.

Tachycardia: width, then regularity

Narrow, regular
SVT (AVNRT, AVRT), flutter 2:1 at about 150, sinus tachycardia

Narrow, irregular
AF, flutter with variable block, multifocal atrial tachycardia

Wide, regular
VT until proven otherwise. SVT with a bundle branch block is less likely.

Wide, irregular
AF with BBB, pre-excited AF (WPW), polymorphic VT or torsades

Unstable patient? Act first and interpret later.

Electrolytes and drugs

High K^+	Peaked T, then flat P and long PR, wide QRS, sine wave
Low K^+	Flat T, ST↓, U waves
High Ca^{2+}	Short QT
Low Ca^{2+}	Long QT
Digoxin	Scooped ST ("reverse tick") and almost any arrhythmia

Don't miss these

- **Wellens:** biphasic or deep T inversion in V2–V3. Critical LAD lesion, often pain-free.
- **de Winter:** upsloping ST↓ with tall T in V1–V6. LAD occlusion.
- **Hyperacute T waves:** broad, tall T waves. Early MI.
- **Pericarditis:** diffuse concave ST↑ with PR↓. Reciprocal in aVR.
- **PE:** sinus tachycardia, S1Q3T3, T inversion in V1–V3.
- **WPW:** short PR with a delta wave.
- **Brugada:** coved ST↑ and T inversion in V1–V2.
- **Long QT:** QTc over 500 ms. Torsades risk.

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